

WONG WAI KUI, CHRIS

Email: kuchrisuk@gmail.com | GitHub: <https://github.com/kuchris> | Website: <https://kuchris.github.io/>

SUMMARY

IT engineer with an MSc in Physics and hands-on experience debugging TCP-based production systems in Japan. Strong in Python, C++, Linux, protocol-level analysis, and scientific computing. Proven ability to identify root causes in multi-layer architectures, improve reliability in networked systems, and build simulation and automation tools with clear technical rigor. Currently based in Japan and available for remote position.

CORE SKILLS

Languages	Python, C++, Julia, JavaScript
Systems	TCP/IP communication, custom binary protocols, client-server debugging, Linux-based development, multi-layer architectures
Backend & Automation	Automation scripting, data pipelines, remote development workflows, Git, Docker
Scientific Computing	Numerical methods, simulation, Monte Carlo methods, data analysis, finite-difference solvers
Libraries & Tools	NumPy, Pandas, SciPy, Matplotlib, PyTorch, CuPy, pySerial, PyVISA, Qt 5, MATLAB, LabVIEW

PROFESSIONAL EXPERIENCE

IT Engineer (SES), ALTEN Japan, Utsunomiya	04/2026 – Present
- Engineer on a railway IC card and cloud ticketing platform spanning host, relay server, and logic server layers. - Analyze and debug TCP-based communication modules built on custom binary protocols in Linux environments. - Identified a critical defect where buffer memory addresses were used as device identifiers, causing unintended disconnects in multi-device scenarios. - Proposed protocol-level device identification, improving connection correctness and strengthening reliability in concurrent device handling. - Support validation, issue analysis, and remote development workflows using Git in a production-style engineering environment.	
Assistant Research Engineer, Amonics Ltd., Hong Kong	12/2021 – 02/2023
- Built Python tools for RS232 communication with laser modules using pySerial, reducing manual testing effort. - Automated GPIB-based measurement collection and processing with PyVISA, improving repeatability of lab workflows. - Performed optical component testing, technical analysis, and hardware issue troubleshooting across multiple instruments. - Produced RMA reports and supported failure investigation with data-backed diagnostics.	

EDUCATION

MANABI Japanese Language Institute Tokyo Campus	01/2025 – 03/2026
MSc in Physics, The Chinese University of Hong Kong	09/2020 – 11/2021
BSc in Applied Physics, The City University of Hong Kong	09/2018 – 07/2020
Higher Diploma in Mechanical Engineering, Hong Kong Institute of Vocational Education	09/2015 – 06/2017

LANGUAGES

English	Fluent (TOEIC 780)
Japanese	JLPT N2 level
Cantonese	Native
Mandarin	Conversational

PROJECTS

Physics Projects

- **2019 – NP-Complete Problems in Combinatorial Optimization:** Applied genetic algorithms and self-organizing maps to solve the traveling salesman problem.
- **2020 – CHSH Inequality and Wave-Particle Duality Demonstration Using Entangled Photon Pairs:** Investigated wave-particle duality, photon entanglement, Bell's theorem, and CHSH inequality using a quED entanglement demonstrator.
- **2021 – Grouped Schrödinger Equation Solvers and Simulations:** Built 1D, 2D, and 3D time-independent solvers plus 1D and 2D time-dependent solvers for arbitrary potentials; includes [1D](#), [2D](#), and [3D](#) TISE outputs.
- **2021 – Classical Analog of Electromagnetically Induced Transparency:** Reproduced EIT-like behavior with coupled RLC circuits as a classical analog of quantum interference effects.
- **2021 – Computing Band Structures:** Computed electronic band structures for materials in Quantum ESPRESSO using density functional theory workflows.
- **2021 – Topological Photonics:** Studied Berry phase, 1D diatomic crystal models, and the integer quantum Hall effect in topological photonics.
- **2022 – Monte Carlo Simulation of the 2D Ising Model in Python:** Simulated thermodynamic properties of a 2D binary alloy model using Monte Carlo methods.
- **2022 – Reproducing EIT-Like Behavior by Solving Coupled Simple Harmonic Oscillators Numerically:** Solved coupled second-order differential equations with multiple ODE solvers.
- **2024 – 2025 – Laser Cooling and Atomic Physics Simulations:** Studied P-wave Feshbach molecules and built simulations for 1D laser cooling and 2D magneto-optical trapping of Rubidium-87; includes [1D cooling](#) and [2D MOT](#).
- **2025 – Numerical Solution of the Gross-Pitaevskii Equation:** Solved the Gross-Pitaevskii equation numerically to analyze Bose-Einstein condensate ground states and dynamics.
- **2025 – Quantum Wavepacket Dynamics in a 2D Lattice with Defects:** Simulated 2D wavepacket propagation with Crank-Nicolson time stepping to study scattering and localization.

Engineering Projects

- **2025 – 1D Schrödinger Equation Solver and Visualizer in C++:** Built an object-oriented C++ desktop application with a separate numerical core and Qt GUI using Eigen and FFTW.
- **2025 – Full-Stack Twitch Clip Audio Extractor with Video Preview and Time Selection:** Developed a web application using Node.js, Express, and FFmpeg for clip download, trimming, and MP3 conversion.
- **2025 – Private Local Retrieval-Augmented Generation (RAG) Pipeline:** Designed a privacy-first local document analysis pipeline using MinerU, Annoy/FAISS, sentence embeddings, and LM Studio.
- **2025 – Reinforcement Learning: Monte Carlo vs Q-Learning in GridWorld:** Implemented and compared Monte Carlo Control and Q-Learning with visualizations of convergence and policy learning.
- **2025 – Realtime Object Detection Using YOLO:** Built a modular real-time object detection system supporting webcam, YouTube, and local video inputs.